Chemistry • Proton transfer, salts and neutralization

Acid Base Chemistry

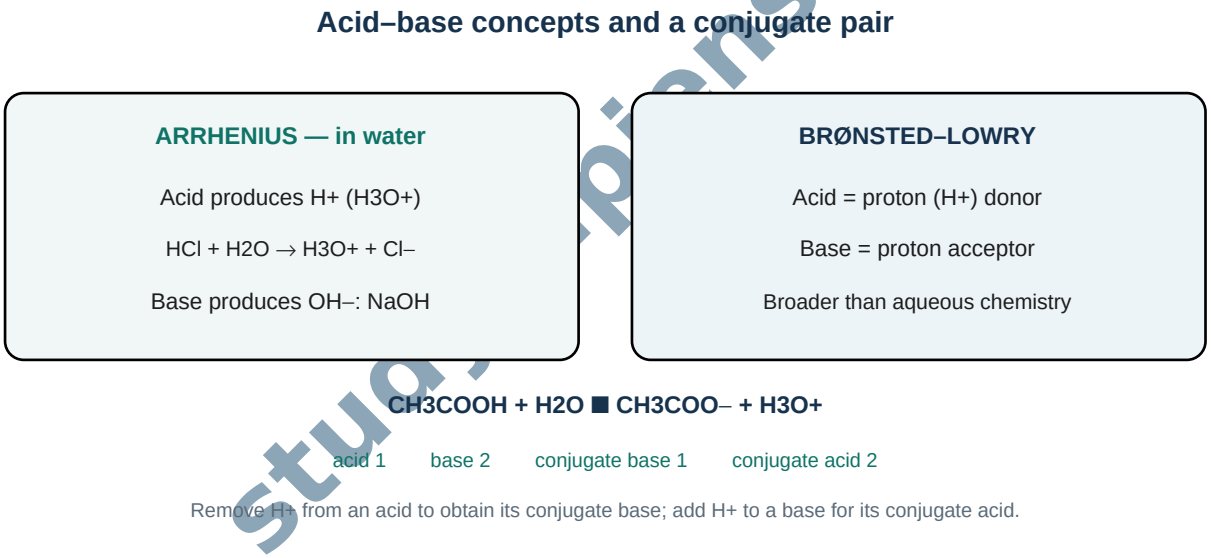
Official exercise • Complete solved answers • Balanced reaction equations

Chapter 7 • Complete official exercise • 10 MCQs + 5 short + 5 constructed + 6 descriptive + 2 investigative

What this chapter is about

• **Arrhenius acids** produce H^+/H_3O^+ in water, while Arrhenius bases produce OH^-.
• **Brønsted–Lowry acids** donate protons and bases accept them. Removing H^+ from an acid gives its conjugate base.
• **Acid reactions** with metals, metal oxides and carbonates produce salts; carbonate reactions also release CO_2.
• **Strength** describes degree of ionization, while **concentration** describes amount dissolved per unit volume. Water and hydrogencarbonate can behave as either acid or base.

Figure 1 — Original comparison of acid–base models and conjugate pairs



Exercise coverage: All five official sections are solved in book order, with balanced equations for each requested reaction.

A. Multiple Choice Questions

1. Which acid is not used as a food or mixed with food?

(a) tartaric acid
(b) ascorbic acid
(c) citric acid
(d) formic acid

Answer: (d) formic acid — Formic acid is corrosive and is not used as a food additive; the other listed acids have food-related uses.

2. While baking, which gas is responsible for raising bread and making it soft?

(a) oxygen
(b) carbon dioxide
(c) nitrogen
(d) carbon monoxide

Answer: (b) carbon dioxide — CO_2 released by fermentation or baking agents forms bubbles that expand the dough.

3. Predict the main characteristics of the reactions of metals with acids.

(a) metals are dissolved
(b) metals are converted into salts
(c) hydrogen gas is evolved
(d) all the above characteristics are true

Answer: (d) all the above characteristics are true — For a sufficiently reactive metal: metal + acid → salt + H_2, so all three statements apply.

4. How many hydroxide ions will calcium hydroxide release in water?

(a) 1
(b) 2
(c) zero
(d) 3

Answer: (b) 2 — $Ca(OH)_2 \rightarrow Ca^{2+} + 2OH^-$; each formula unit supplies two hydroxide ions.

5. In neutralization between KOH and H_3PO_4, how many molecules of KOH react with one molecule of H_3PO_4?

(a) 2
(b) 1
(c) 3
(d) 4

Answer: (c) 3 — $3KOH + H_3PO_4 \rightarrow K_3PO_4 + 3H_2O$, so three KOH formula units are required.

6. Which acid is used in the preparation of soap?

(a) tartaric acid
(b) citric acid
(c) stearic acid
(d) oxalic acid

Answer: (c) stearic acid — Stearic acid and other long-chain fatty acids form sodium or potassium salts called soaps.

7. Which compound is formed when SO_3 is dissolved in water?

(a) SO_3
(b) H_2SO_3
(c) H_2SO_4
(d) $H_2S_2O_7$

Answer: (c) H_2SO_4 — $SO_3 + H_2O \rightarrow H_2SO_4$, sulphuric acid.

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8. Which of the following contains oxalic acid?

(a) tomato
(b) orange
(c) tamarind
(d) sour milk

Answer: (a) tomato — Tomato contains oxalic acid; orange contains citric acid, tamarind tartaric acid and sour milk lactic acid.

9. Which compound behaves as a conjugate base in $CH_3COOH(aq) + H_2O(l) \rightleftharpoons CH_3COO^-(aq) + H_3O^+(aq)$?

(a) CH_3COOH
(b) H_2O
(c) CH_3COO^-
(d) H_3O^+

Answer: (c) CH_3COO^- — CH_3COOH donates H^+ and becomes CH_3COO^-, its conjugate base.

10. A reaction with substance Z produces a gas that turns red litmus paper blue. What is the reaction?

(a) reaction of an acid with a metal carbonate
(b) reaction of an acid with a metal hydrogen carbonate
(c) reaction of an alkali with an acid
(d) reaction of an alkali with an ammonium salt

Answer: (d) reaction of an alkali with an ammonium salt — Alkali + ammonium salt releases NH_3, an alkaline gas that turns moist red litmus blue.

B. Questions for Short Answers

7.1. Choose Arrhenius acids among HF, NH_3, H_2SO_4, SO_3, H_2S and H_2O.

The clear Arrhenius acids are **HF, H_2SO_4 and H_2S** because they donate H^+ in aqueous solution:

$$HF \rightleftharpoons H^+ + F^-$$

$$H_2SO_4 \rightarrow 2H^+ + SO_4^{2-} \text{ (complete for the first proton)}$$

$$H_2S \rightleftharpoons H^+ + HS^-$$

NH_3 is a base, and SO_3 is an acidic oxide that must first react with water to form H_2SO_4. Water is **amphiprotic**: it can act as an Arrhenius acid by producing H^+ or a base by producing OH^-, so it is not placed only in the acid list.

7.2. How does calcium metal react with dilute H_2SO_4?

$$Ca(s) + H_2SO_4(aq) \rightarrow CaSO_4(s/aq) + H_2(g)\uparrow$$

Calcium initially reacts vigorously, producing hydrogen bubbles and calcium sulphate. $CaSO_4$ is only sparingly soluble, so a coating can form on the metal and slow further reaction. Calcium also reacts with water; this is not a safe casual

classroom experiment.

7.3. Which salt is formed when HCl reacts with $BaCO_3$?

The salt is **barium chloride, $BaCl_2$**: $BaCO_3 + 2HCl \rightarrow BaCl_2 + H_2O + CO_2\uparrow$.

7.4. How will you justify that HSO_4^- is a Brønsted–Lowry acid?

HSO_4^- can donate a proton, which is the defining property of a Brønsted–Lowry acid: $HSO_4^- + H_2O \rightleftharpoons SO_4^{2-} + H_3O^+$. It is amphiprotic because it can also accept H^+ to form H_2SO_4.

7.5. Why is HCl not edible although it is present in the stomach and responsible for digestion of food?

Gastric juice contains HCl at a controlled, very dilute concentration together with protective mucus and other secretions. It activates pepsinogen and helps kill microbes. Laboratory or concentrated HCl is highly corrosive and would burn the mouth, oesophagus and stomach; even the stomach lining can be damaged when its protective control fails. A natural physiological role does not make the pure chemical safe to consume.

C. Constructed Response Questions

7.1. What chemical name will you give to soap as a compound?

Soap is a **sodium or potassium salt of a long-chain fatty acid**. Examples are sodium stearate, $C_{17}H_{35}COONa$, and sodium palmitate, $C_{15}H_{31}COONa$. Commercial soap is usually a mixture rather than one pure compound.

7.2. Water ionizes in the presence of acid. Should water be called an acid, a base or both?

Water is **both acid and base (amphiprotic)**. It can accept a proton from an acid or donate a proton to a base:

$$\text{With acid: } HCl + H_2O \rightarrow H_3O^+ + Cl^- \text{ (water = base)}$$

$$\text{With base: } NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^- \text{ (water = acid)}$$

$$\text{Autoionization: } 2H_2O \rightleftharpoons H_3O^+ + OH^-$$

The simplified expression $H_2O \rightleftharpoons H^+ + OH^-$ conveys the same ion formation, but free H^+ is hydrated in water.

7.3. Why does Na_2CO_3 behave like a base in water?

Sodium carbonate dissolves to give Na^+ and CO_3^{2-}. Carbonate is the conjugate base of the weak acid HCO_3^- and accepts a proton from water:

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$$CO_3^{2-} + H_2O \rightleftharpoons HCO_3^- + OH^-$$

The OH^- produced makes the solution alkaline. Na^+ is a spectator and does not neutralize this hydrolysis.

7.4. Is $NaHCO_3$ a base or an acid? Justify your answer.

Sodium hydrogencarbonate is **amphiprotic**, although it commonly acts as a mild base and its solution is weakly alkaline.

$$\text{As base: } HCO_3^- + H^+ \rightarrow H_2CO_3 \rightarrow CO_2 + H_2O$$

$$\text{As acid: } HCO_3^- + OH^- \rightarrow CO_3^{2-} + H_2O$$

Na^+ is a spectator. HCO_3^- can accept H^+ from an acid or donate H^+ to a stronger base, so its role depends on the other reactant.

7.5. What is the difference between a strong acid and a concentrated acid?

Term	Meaning	Independent quantity
Strong acid	Ionizes almost completely in water	Strength (degree of ionization)
Concentrated acid	Contains a relatively large amount of acid per unit volume	Concentration (mol dm^{-3} or related measure)

A strong acid can be dilute, for example 0.001 M HCl, and a weak acid can be concentrated, for example concentrated CH_3COOH. “Strong” does not mean “concentrated,” and dilution does not change an acid’s intrinsic strength.

D. Descriptive Questions

7.1. Explain the Arrhenius concept of acids and bases.

Arrhenius acid: a substance that increases H^+ concentration in water; hydrated H^+ is represented more accurately as H_3O^+. **Arrhenius base:** a substance that increases OH^- concentration in water.

$$HCl + H_2O \rightarrow H_3O^+ + Cl^-$$

$$NaOH(aq) \rightarrow Na^+(aq) + OH^-(aq)$$

$$\text{Neutralization: } H^+ + OH^- \rightarrow H_2O$$

Strong electrolytes ionize extensively; weak ones ionize partially. The model explains aqueous neutralization but is limited: it requires water and does not directly explain basic NH_3, proton transfer in non-aqueous solvents or acidic oxides that first react with water.

7.2. Compare Arrhenius and Brønsted–Lowry concepts of acids and bases.

Feature	Arrhenius concept	Brønsted–Lowry concept
Acid	Produces H^+/H_3O^+ in water	Donates a proton H^+
Base	Produces OH^- in water	Accepts a proton H^+
Reaction focus	Ion formation in aqueous solution	Proton transfer between conjugate pairs
Medium	Water only	Aqueous or non-aqueous; also gas phase
NH_3	Explained indirectly because aqueous NH_3 forms OH^-	Directly a base: $NH_3 + H^+ \rightarrow NH_4^+$
Conjugate pairs	Not central	Central: acid/base differ by one H^+

Brønsted–Lowry is broader and includes many Arrhenius cases, but both remain models with specific uses.

7.3. How does sulphuric acid react with NH_4Cl, NH_4NO_3, MgO and $MgCO_3$?

Reactant	Balanced equation	Observation / type
NH_4Cl	$2NH_4Cl + H_2SO_4 \rightarrow (NH_4)_2SO_4 + 2HCl$	With concentrated acid and heating, volatile HCl is displaced.
NH_4NO_3	$2NH_4NO_3 + H_2SO_4 \rightarrow (NH_4)_2SO_4 + 2HNO_3$	Concentrated acid can displace the more volatile HNO_3.
MgO	$MgO + H_2SO_4 \rightarrow MgSO_4 + H_2O$	Metal oxide + acid → salt + water.
$MgCO_3$	$MgCO_3 + H_2SO_4 \rightarrow MgSO_4 + H_2O + CO_2\uparrow$	Carbonate + acid → salt + water + CO_2.

The first two equations describe the concentrated-acid/heating displacement reactions. In dilute aqueous solution, all species are ionic and there is no comparable simple precipitation or gas-driving reaction. Concentrated H_2SO_4 and ammonium-nitrate mixtures are hazardous and are not casual classroom experiments.

7.4. What happens when a base reacts with a non-metallic oxide? What do you infer about the nature of the non-metallic oxide?

A base reacts with an **acidic non-metallic oxide** to form a salt and usually water.

$$2NaOH + CO_2 \rightarrow Na_2CO_3 + H_2O$$

$$Ca(OH)_2 + SO_2 \rightarrow CaSO_3 + H_2O$$

The reaction shows that these non-metallic oxides have **acidic character**: they neutralize bases and often react with water to form acids. They are called acidic oxides or acid anhydrides. Not every non-metallic oxide behaves this way; CO, NO and N_2O are commonly classed as neutral oxides.

7.5. State the reason for showing acidic character by both dry HCl gas and HCl solution in water.

Both contain hydrogen chloride, a covalent molecule capable of donating a proton to a suitable base. Dry HCl gas can react directly with a base such as NH_3: $HCl(g) + NH_3(g) \rightarrow NH_4Cl(s)$, so it has Brønsted acidic character.

However, dry HCl gas is not ionized into H_3O^+ and does not show the usual aqueous-acid tests, such as changing dry litmus or conducting as an aqueous electrolyte does.

$$\text{In water: } HCl + H_2O \rightarrow H_3O^+ + Cl^-$$

Aqueous HCl shows characteristic acid behaviour because water accepts the proton and mobile H_3O^+ and Cl^- ions form. Moisture is therefore required for ordinary litmus tests.

7.6. Differentiate between an acid and its conjugate base.

They are related by the loss of exactly one proton. The acid has one more H^+ and one unit more positive charge than its conjugate base.

Acid (before donation)	Conjugate base (after H^+ donation)
HCl	Cl^-
CH_3COOH	CH_3COO^-
HSO_4^-	SO_4^{2-}
H_2O	OH^-
NH_4^+	NH_3

$$CH_3COOH + H_2O \rightleftharpoons CH_3COO^- + H_3O^+$$

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The acid is the proton donor in the forward reaction; its conjugate base can accept that proton in the reverse reaction. A strong acid has a very weak conjugate base, whereas a weak acid has a comparatively stronger conjugate base.

E. Investigative Questions

7.1. Acids play significant roles within the human body. Comment on this statement.

The statement is correct: many acids are essential when their concentrations and locations are controlled.

• **Hydrochloric acid:** Gastric HCl activates pepsinogen, aids protein digestion and limits many ingested microbes.

• **Carbonic acid / bicarbonate system:** $CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$ helps buffer blood pH and transport CO_2.

• **Lactic acid:** It can form during intense anaerobic muscle activity and contributes to the body's energy metabolism.

• **Amino, fatty and nucleic acids:** These are essential for proteins, membranes/energy and genetic information respectively.

• **Ascorbic acid:** Vitamin C supports collagen formation and antioxidant defence.

Control is crucial. Acid reflux, excessive gastric acidity or severe metabolic acidosis can injure tissues; essential biological acids are not therefore safe as concentrated laboratory chemicals.

7.2. What is observed when CO_2 is passed through lime water (i) for a short duration and (ii) for a long duration?

1. Short duration: Lime water turns milky because insoluble white calcium carbonate precipitates.

2. Long duration / excess CO_2: The milkiness gradually disappears because the precipitate reacts further and forms soluble calcium hydrogencarbonate.

$$\text{(i) } Ca(OH)_2(aq) + CO_2(g) \rightarrow CaCO_3(s)\downarrow + H_2O(l)$$

$$\text{(ii) } CaCO_3(s) + CO_2(g) + H_2O(l) \rightarrow Ca(HCO_3)_2(aq)$$

Thus the sequence is **clear → milky → clear again**, provided excess CO_2 is passed.

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Points to Remember

Arrhenius: acid gives H^+/H_3O^+ and base gives OH^- in water.

Brønsted–Lowry: acid donates H^+, base accepts H^+; conjugate partners differ by one proton.

Acid + reactive metal → salt + H_2; acid + metal oxide → salt + water; acid + carbonate → salt + water + CO_2.

Acid + base → salt + water in the usual aqueous neutralization.

Strong means extensively ionized; **concentrated** means much solute per unit volume.

Water, HSO_4^- and HCO_3^- are amphiprotic: their acid/base role depends on the partner.

Dry HCl is covalent; aqueous HCl forms H_3O^+ and Cl^- and shows typical electrolytic acid tests.

Short CO_2 treatment makes lime water milky; excess CO_2 makes it clear through soluble $Ca(HCO_3)_2$.

MCQ key at a glance

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	d	2	b	3	d	4	b	5	c
6	c	7	c	8	a	9	c	10	d

How to attempt this chapter in the paper

- Balance the salt formula before balancing the full acid reaction, especially with di- and trivalent ions.
- For conjugate pairs, remove exactly one H^+ and lower the charge by one positive unit.
- In an explanation, name the ion responsible for a test: H_3O^+, OH^-, CO_2, NH_3 or $CaCO_3$.
- Do not use “strong” and “concentrated” interchangeably, and never infer edibility from biological presence.